

Phishing Email Detection & Awareness Report

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Internship: Future Interns – Cyber Security Track **Tools Used:** Nazario Phishing Corpus (public dataset), manual header/link analysis

Executive Summary

This report analyzes four real-world phishing email samples to identify common attack patterns, classify their risk level, and provide practical prevention guidance. The goal is to help end users recognize phishing attempts before they cause harm.

Case 1: Fake cPanel Account Verification

Sender: service@cpanel.com (spoofed) **Subject:** Verify Your Account **Classification:** Phishing

Indicators Identified:

Generic greeting ("Dear client") instead of a personalized name

False urgency, framing account re-verification as mandatory

Link text displayed as a trusted-looking domain, but actual destination cannot be verified without hovering a classic spoofing technique

Risk: Medium-High — could lead to credential theft if the user enters login details on a fake page.

Case 2: Fake IT Help Desk Password Expiry

Sender: Impersonated internal employee address **Subject:** IT-Service Help Desk **Classification:** Phishing

Indicators Identified:

Sender name does not match a legitimate internal IT department address

Message creates false urgency with a fake "password expires in 3 days" warning

Vague "Click Here" call-to-action with no visible destination a common way to hide malicious links

Risk: High — targets credential harvesting by mimicking routine internal IT communication, which employees are conditioned to trust.

Case 3: Fake PayPal Account Update Request

Sender: infos@2015p.com (impersonating PayPal) **Subject:** Dear Client: Update Your Information **Classification:** Phishing

Indicators Identified:

Sender domain ("2015p.com") has no relation to PayPal's real domain (paypal.com)

Generic greeting ("Dear Client") rather than the user's actual name

Vague claim that account information is "inaccurate or unverified," pressuring the user to click through

Risk: High — brand impersonation of a trusted financial service increases the likelihood of successful credential theft.

Case 4: Fake Storage Limit Warning

Sender: mail@admin.com **Subject:** Administrator **Classification:** Phishing

Indicators Identified:

Sender address is generic and not tied to any specific real organization

Uses a fabricated storage-limit scare tactic ("account exceeded its limit")

24-hour deadline creates artificial urgency to bypass careful judgment

Risk: Medium — less sophisticated, but still effective against less security-aware users.

Common Patterns Across All Samples

Pattern	Present In
Generic/impersonal greeting	Cases 1, 3
False urgency / deadline	Cases 1, 2, 4
Sender domain mismatch	Cases 1, 3, 4
Vague "Click Here" links	Cases 2, 4

Prevention & Awareness Guidelines

1. **Check the sender's actual email address**, not just the display name — hover over or tap to reveal the full address.
2. **Be suspicious of urgency** — legitimate organizations rarely demand immediate action within hours.
3. **Hover over links before clicking** to preview the actual destination URL.
4. **Verify independently** — if an email claims to be from your bank, IT department, or a service like PayPal, log in directly through the official website or app instead of clicking the email link.
5. **Watch for generic greetings** like "Dear Client" or "Dear User" instead of your actual name — legitimate services usually personalize emails.

6. **Report suspicious emails** to your IT/security team rather than deleting them silently, so others can be warned.
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Conclusion

All four analyzed samples exhibited multiple classic phishing indicators — spoofed sender domains, false urgency, and generic language designed to bypass user scrutiny. Recognizing these patterns is the first line of defense against phishing attacks, and awareness training remains one of the most effective mitigations alongside technical email filtering.
